



Mathematics Extension 2

General Instructions

- Working time: 1½ hours
- Reading time: 5 minutes
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- A separate section with graphs is supplied for Question 2 responses.

Total marks – 72

- Attempt Questions 1 – 3
- All questions are of equal value
- Begin each question in a new booklet
- All necessary working should be shown in every question.

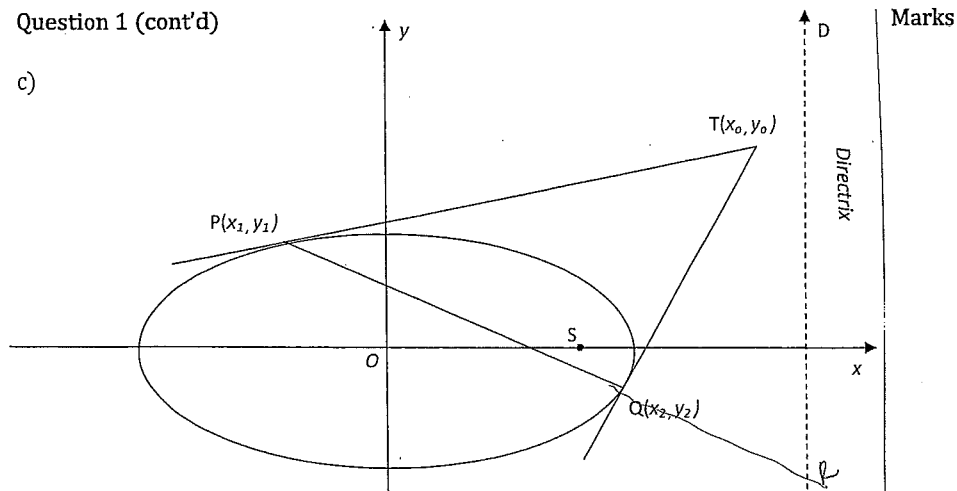
Question 1 – (24 marks) – (Start a new booklet)

Marks

- a) Find the equation of the ellipse which has eccentricity $\frac{3}{4}$ and foci at $S'(-3, 0)$ and $S(3, 0)$ 2
- b) Consider the hyperbola H with equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$
- (i) Find the eccentricity, e 1
- (ii) Find the coordinates of the foci, S and S' 1
- (iii) Find the equation of each directrix 1
- (iv) Find the equation of each of the asymptotes 1
- (v) Sketch the hyperbola H , showing all the main features. 2
- (vi) Given $P(4 \sec \phi, 3 \tan \phi)$ is any arbitrary point on H
- (α) Show that the equation of the tangent at P is $\frac{x \sec \phi}{4} - \frac{y \tan \phi}{3} = 1$ 3
- (β) The tangent at P meets the asymptote of the hyperbola H at the points Q and Q' . Find the coordinates of Q and Q' 2
- (γ) Hence, or otherwise, show that $PQ = PQ'$ 2

Question 1 (cont'd)

c)



The ellipse E has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with focus S and directrix D as shown on the diagram. The point $T(x_0, y_0)$ lies outside the ellipse and is not on the x -axis.

The chord of contact PQ from T intersects the directrix at R .

- (i) Show that the equation of the tangent to ellipse at the point $P(x_1, y_1)$ is 2

$$\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1$$

- (ii) Show that the equation of the normal to the ellipse at the point $P(x_1, y_1)$ is 2

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

- (iii) Given the equation of the chord of contact from T is $\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$, show that $R(x, y)$ has ordinate 2

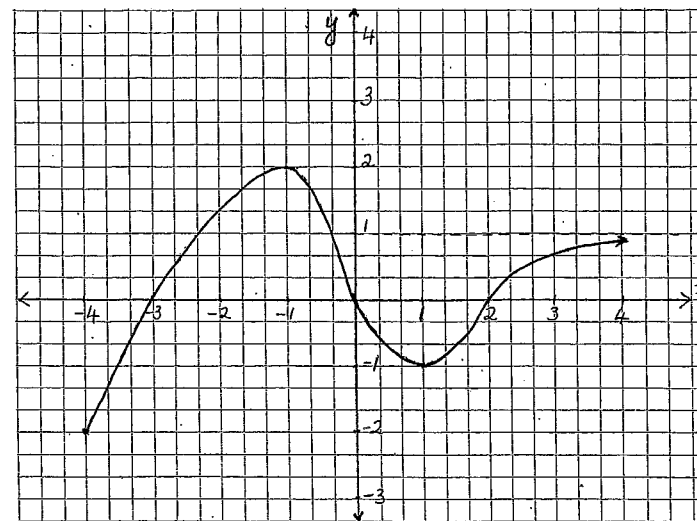
$$y = \frac{b^2}{y_0} \left(1 - \frac{x_0}{ae} \right)$$

- (iv) Show that TS is perpendicular to SR . 3

Question 2 – (24 marks) – (Start a new booklet)

Marks

- a) The graph of $y = f(x)$ for $x \geq -4$ is shown in the following diagram.



On the separate diagrams provided, neatly sketch the following graphs.

- (i) $y = f(x + 1)$ 2

- (ii) $y = [f(x)]^2$ 2

- (iii) $y = \sqrt{f(x)}$ 2

- (iv) $y^2 = f(x)$ 2

- (v) $y = f(|x|)$ 2

- (vi) $y = \frac{1}{f(x)}$ 2

- (vii) $y = f'(x)$ 2

Question 2 (cont'd)

Marks

b) For the curve $2x^2 - xy + y^2 = 14$

(i) Use implicit differentiation to find $\frac{dy}{dx}$

3

(ii) Find the gradient of the tangent at the point $(1, -3)$

1

(iii) Show that any stationary points must lie on the line $y = 4x$

1

(iv) Hence find the coordinates of the stationary points

3

c) Sketch the graph of $\cos(x + y) = 0$ $-3\pi \leq x \leq 3\pi$

2

Question 3 – (24 marks) – (Start a new booklet)

Marks

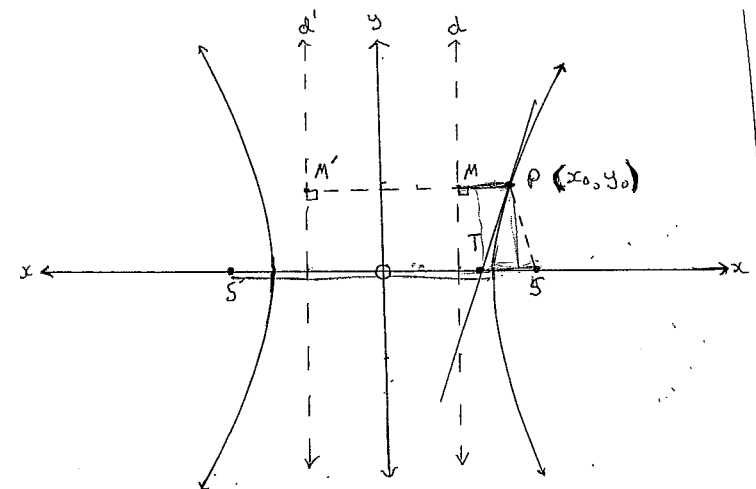
a) (i) On separate diagrams draw neat sketches of $y = \cos x$ for $-2\pi \leq x \leq 2\pi$ and $y = 3^x$ for $-1 \leq x \leq 1$

2

(ii) Using part (i) or otherwise draw a neat sketch of $y = 3^{\cos x}$ $-2\pi \leq x \leq 2\pi$

2

b)



The point $P(x_0, y_0)$ lies on the hyperbola $\frac{x^2}{9} - \frac{y^2}{7} = 1$

The tangent to the hyperbola at P cuts the x -axis at T , and has equation $\frac{x_0 x}{9} - \frac{y_0 y}{7} = 1$. The two foci of the hyperbola are S and S' , and the two directrices are d and d' .

The points M and M' are the feet of the perpendicular from P to d and d' .

The eccentricity is $\frac{4}{3}$, the foci $S(4, 0)$ and $S'(-4, 0)$, the directrices are $x = \pm \frac{9}{4}$

(i) Show that T has coordinates $(\frac{9}{x_0}, 0)$

1

(ii) Using the focus directrix definition, or otherwise, show that $\frac{PS}{PS'} = \frac{TS}{TS'}$

3

(iii) Prove that $|PS' - PS| = 6$

1

Question 3 (cont'd)

Marks

- c) Show that, if $y = mx + k$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then $m^2 a^2 - b^2 = k^2$.

3

- d) (i) Draw a neat sketch of the parabola $y = x^2 - 2x - 3$. Your sketch should clearly show the intercepts with the coordinate axes and the coordinates of the vertex.

3

- (ii) Hence, or otherwise, sketch the graph of $y = 4 - |x^2 - 2x - 3|$ showing all important features.

2

- (iii) For what values of m does the equation $mx = 4 - |x^2 - 2x - 3|$ have 4 distinct real solutions?

3

- e) P is any point on the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and S is a focus of E .

Prove that the line through S perpendicular to the tangent at P meets the line OP produced at a point on the directrix.

4

You may use the fact that the equation of the tangent through P is

$$y = \frac{-b \cos \theta}{a \sin \theta} \left(x - \frac{a}{\cos \theta} \right)$$

Year 12 Mid-Course Assessment Ext 2 Maths

Question 1.

a) $e = \frac{3}{4}$, $S(3,0) \therefore ae = 3$
 $a \times \frac{3}{4} = 3$
 $\therefore a = 4$
 Now $b^2 = a^2(1 - e^2)$
 $b^2 = 16(1 - \frac{9}{16})$
 $= 16 - 9$
 $= 7$
 $\therefore$ Equation of ellipse is $\frac{x^2}{16} + \frac{y^2}{7} = 1$

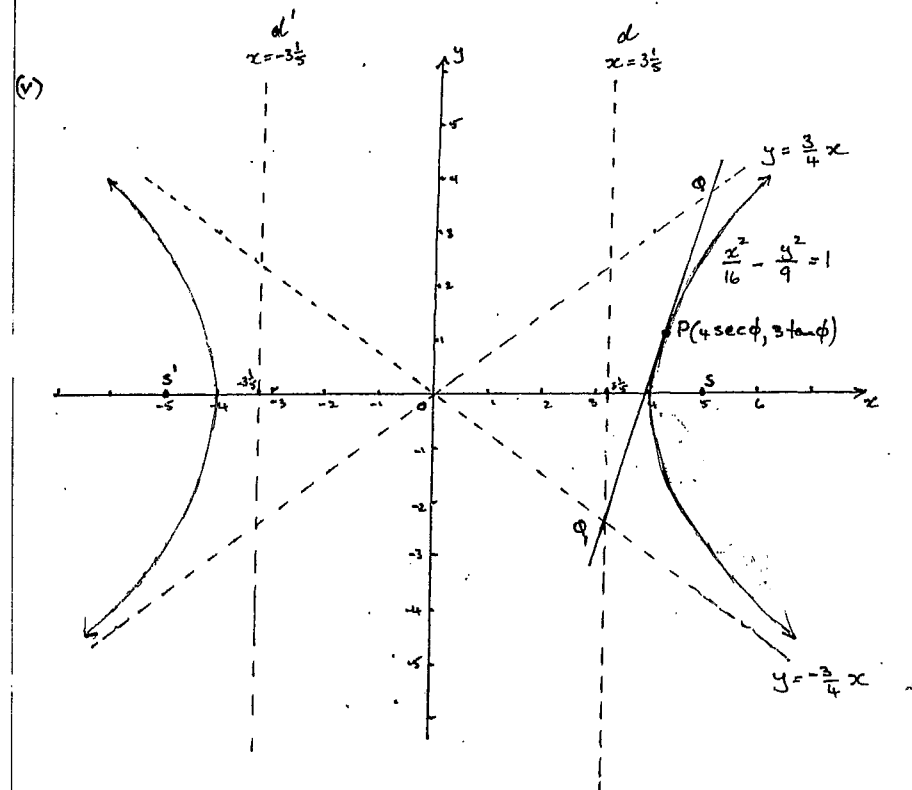
b) (i) Hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ has $a = 4$, $b = 3$

Since $b^2 = a^2(e^2 - 1)$
 $9 = 16(e^2 - 1)$
 $\frac{9}{16} + 1 = e^2$
 $\frac{25}{16} = e^2$
 $\therefore e = \frac{5}{4}$, $e > 0$

(ii) Foci are $S(ae, 0)$ i.e. $S(4 \times \frac{5}{4}, 0)$; $S(5, 0)$
 $S'(-5, 0)$

(iii) Directrices are $x = \pm \frac{a}{e}$
 i.e. $x = \pm \frac{4}{5/4}$
 i.e. $x = \frac{16}{5}$ or $x = -3\frac{1}{5}$

(iv) Asymptotes are $y = \pm \frac{b}{a}x$
 i.e. $y = \frac{3}{4}x$ and $y = -\frac{3}{4}x$



(vi) $P(4 \sec \phi, 3 \tan \phi)$

(a) $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$\frac{2x}{16} - \frac{2y}{9} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x}{16} \cdot \frac{-9}{2y}$

$= \frac{9x}{16y} = \frac{9 \times 4 \sec \phi}{16 \times 3 \tan \phi}$, when at P.

$\therefore$ Equation of tangent is

$y - 3 \tan \phi = \frac{3 \sec \phi}{4 \tan \phi} (x - 4 \sec \phi)$

$4y \tan \phi - 12 \tan^2 \phi = 3x \sec \phi - 12 \sec^2 \phi$

$3x \sec \phi - 4y \tan \phi = 12(\sec^2 \phi - \tan^2 \phi)$

$3x \sec \phi - 4y \tan \phi = 12$, $\tan^2 \phi + 1 = \sec^2 \phi$

$\div 12, \therefore \frac{x \sec \phi}{4} - \frac{y \tan \phi}{3} = 1$

p) tangent at P meets asymptotes at $\Phi + \Phi'$

ie $y = \pm \frac{3}{4}x$ — (1)

$\frac{x \sec \phi}{4} - \frac{y \tan \phi}{3} = 1$ — (2)

Sub (1) into 2: $\frac{x \sec \phi}{4} - \frac{\pm 3x \tan \phi}{4 \times 3} = 1$

$\frac{x}{4} (\sec \phi \mp \tan \phi) = 1$

ie $\begin{cases} x = \frac{4}{\sec \phi \mp \tan \phi} \\ y = \pm \frac{3}{4} \times \frac{4}{\sec \phi \mp \tan \phi} \end{cases}$

$\therefore Q \left(\frac{4}{\sec \phi - \tan \phi}, \frac{3}{\sec \phi - \tan \phi} \right) + Q' \left(\frac{4}{\sec \phi + \tan \phi}, \frac{-3}{\sec \phi + \tan \phi} \right)$

8) Now $PQ^2 = \left(4 \sec \phi - \frac{4}{\sec \phi - \tan \phi} \right)^2 + \left(3 \tan \phi - \frac{3}{\sec \phi - \tan \phi} \right)^2$

$= 16 \left(\frac{\sec^2 \phi - \sec \phi \tan \phi - 1}{\sec \phi - \tan \phi} \right)^2 + 9 \left(\frac{\sec \phi \tan \phi - \tan^2 \phi - 1}{\sec \phi - \tan \phi} \right)^2$

$= 16 \left(\frac{\tan^2 \phi - \sec \phi \tan \phi}{\sec \phi - \tan \phi} \right)^2 + 9 \left(\frac{\sec \phi \tan \phi - \sec^2 \phi}{\sec \phi - \tan \phi} \right)^2, \tan^2 \phi + 1 = \sec^2 \phi$

$= 16 \tan^2 \phi \left(\frac{\tan \phi - \sec \phi}{\sec \phi - \tan \phi} \right)^2 + 9 \sec^2 \phi \left(\frac{\tan \phi - \sec \phi}{\sec \phi - \tan \phi} \right)^2$

$= 16 \tan^2 \phi + 9 \sec^2 \phi$

Also, $(PQ')^2 = \left(4 \sec \phi - \frac{4}{\sec \phi + \tan \phi} \right)^2 + \left(3 \tan \phi - \frac{-3}{\sec \phi + \tan \phi} \right)^2$

$= 16 \left(\frac{\sec^2 \phi + \sec \phi \tan \phi - 1}{\sec \phi + \tan \phi} \right)^2 + 9 \left(\frac{\sec \phi \tan \phi + \tan^2 \phi + 1}{\sec \phi + \tan \phi} \right)^2$

$= 16 \left(\frac{\tan^2 \phi + \sec \phi \tan \phi}{\sec \phi + \tan \phi} \right)^2 + 9 \left(\frac{\sec \phi \tan \phi + \sec^2 \phi}{\sec \phi + \tan \phi} \right)^2, \tan^2 \phi + 1 = \sec^2 \phi$

$= 16 \tan^2 \phi \left(\frac{\tan \phi + \sec \phi}{\sec \phi + \tan \phi} \right)^2 + 9 \sec^2 \phi \left(\frac{\tan \phi + \sec \phi}{\sec \phi + \tan \phi} \right)^2$

$= 16 \tan^2 \phi + 9 \sec^2 \phi$

$\therefore$ Since $(PQ)^2 = (PQ')^2$

then,

$PQ = PQ'$

OR ALTERNATIVELY and more SIMPLY!

(8)

Midpoint of QQ' is:

$M \left[\frac{4}{2} \left(\frac{1}{\sec \phi - \tan \phi} + \frac{1}{\sec \phi + \tan \phi} \right), \frac{1}{2} \left(\frac{3}{\sec \phi - \tan \phi} + \frac{-3}{\sec \phi + \tan \phi} \right) \right]$

ie $M \left[\frac{4}{2} \left(\frac{\sec \phi + \tan \phi + \sec \phi - \tan \phi}{\sec^2 \phi - \tan^2 \phi} \right), \frac{3}{2} \left(\frac{\sec \phi + \tan \phi - \sec \phi + \tan \phi}{\sec^2 \phi - \tan^2 \phi} \right) \right]$

ie $M \left[2 \left(\frac{2 \sec \phi}{1} \right), \frac{3}{2} \left(\frac{2 \tan \phi}{1} \right) \right]$

$\therefore \sin^2 \phi + \cos^2 \phi = 1$

$\tan^2 \phi + 1 = \sec^2 \phi$

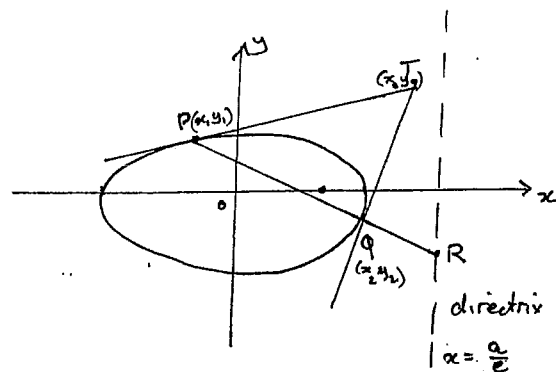
Thus midpoint of QQ' is

$(4 \sec \phi, 3 \tan \phi)$

But This is point P

Thus, $PQ = PQ'$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



$$(i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y} = -\frac{b^2 x_1}{a^2 y_1} \text{ when at } P(x_1, y_1)$$

$\therefore$ Equation of tangent at P is

$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y_1 y - a^2 y_1^2 = -b^2 x_1 x + b^2 x_1^2$$

$$b^2 x_1 x + a^2 y_1 y = a^2 y_1^2 + b^2 x_1^2$$

$$\div a^2 b^2; \quad \frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$$\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1 \quad \text{since } (x_1, y_1) \text{ lies on } E.$$

(ii) Gradient of normal is $\frac{a^2 y_1}{b^2 x_1}$

$\therefore$ Equation of normal is

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$b^2 x_1 y - b^2 x_1 y_1 = a^2 y_1 x - a^2 x_1 y_1$$

$$a^2 y_1 x - b^2 x_1 y = a^2 x_1 y_1 - b^2 x_1 y_1$$

$$\div x_1 y_1; \quad \frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(iii) Equation of chord from T is

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$$

Ordinate of R when $x = \frac{a}{e}$

$$\text{ie } \frac{x_0 \times a}{a^2 e} + \frac{y_0 y}{b^2} = 1$$

$$\frac{x_0}{ae} + \frac{y_0 y}{b^2} = 1$$

$$\frac{y_0 y}{b^2} = 1 - \frac{x_0}{ae}$$

$$y_0 y = b^2 \left(1 - \frac{x_0}{ae}\right)$$

$$y = \frac{b^2}{y_0} \left(1 - \frac{x_0}{ae}\right)$$

(iv) Now,

$$m_{TS} = \frac{y_0 - 0}{x_0 - ae}$$

$$= \frac{y_0}{x_0 - ae}$$

$$m_{SR} = \frac{\frac{b^2}{y_0} \left(1 - \frac{x_0}{ae}\right) - 0}{\frac{a}{e} - ae}$$

$$= \frac{b^2 \left(1 - \frac{x_0}{ae}\right) \times ae}{ay_0 \left(\frac{1}{e} - e\right) \times ae}$$

$$= \frac{b^2 (ae - x_0)}{a^2 y_0 (1 - e^2)}$$

$$= \frac{b^2 (ae - x_0)}{b^2 y_0} \quad b^2 = a^2 (1 - e^2)$$

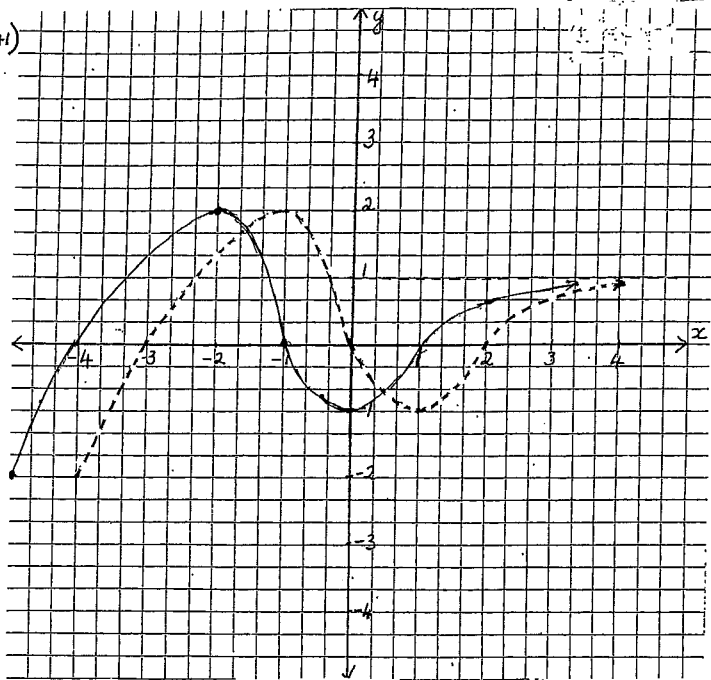
$$= \frac{ae - x_0}{y_0}$$

$$\text{Since } m_{TS} \cdot m_{SR} = \frac{y_0}{x_0 - ae} \cdot \frac{ae - x_0}{y_0}$$

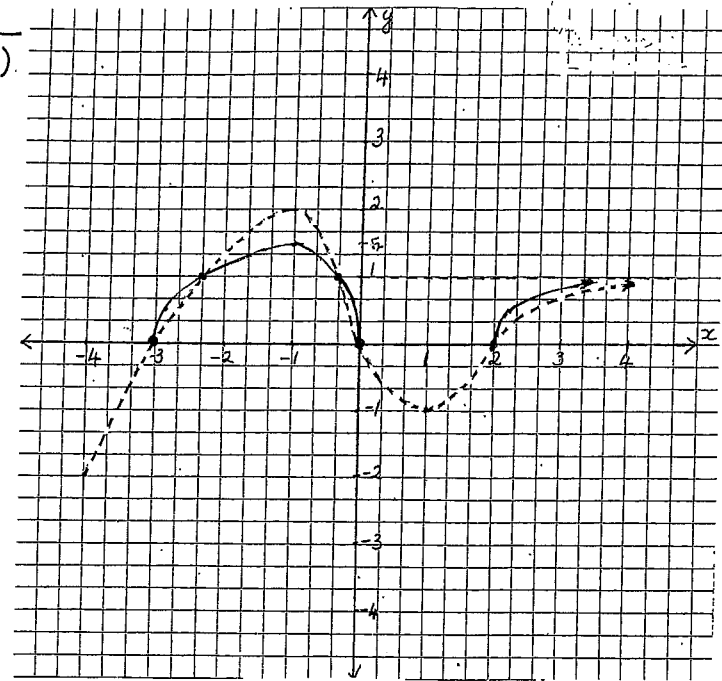
$$= -1 \quad \text{then, } TS \perp SR$$

Question 2 (a)

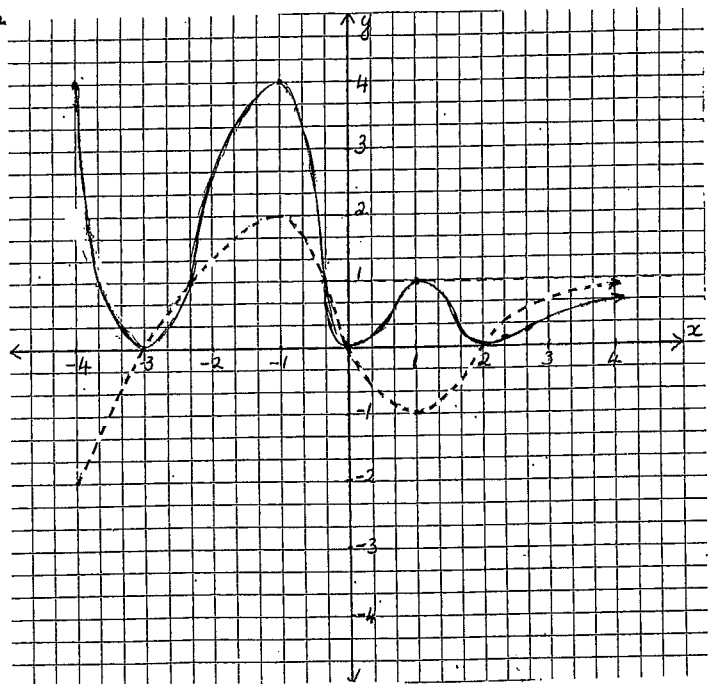
(i) $y = f(x+1)$



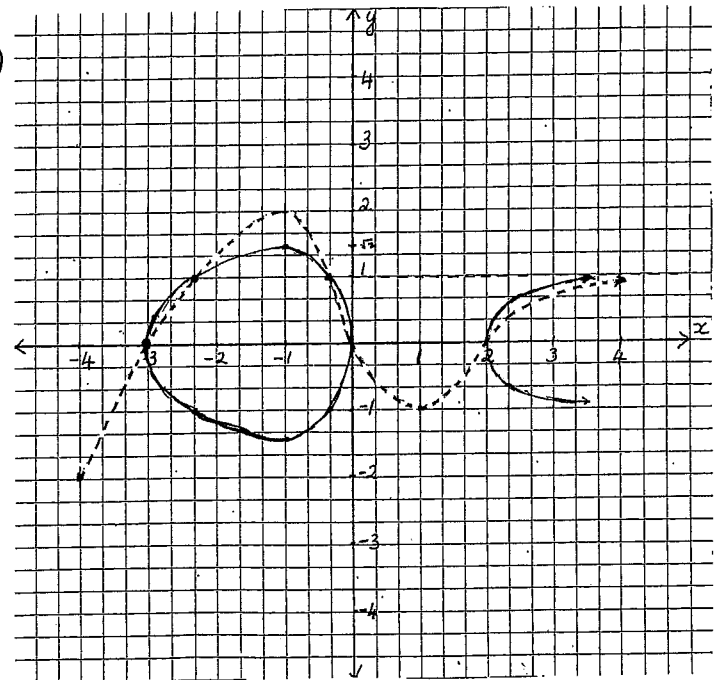
(iii) $y = \sqrt{f(x)}$



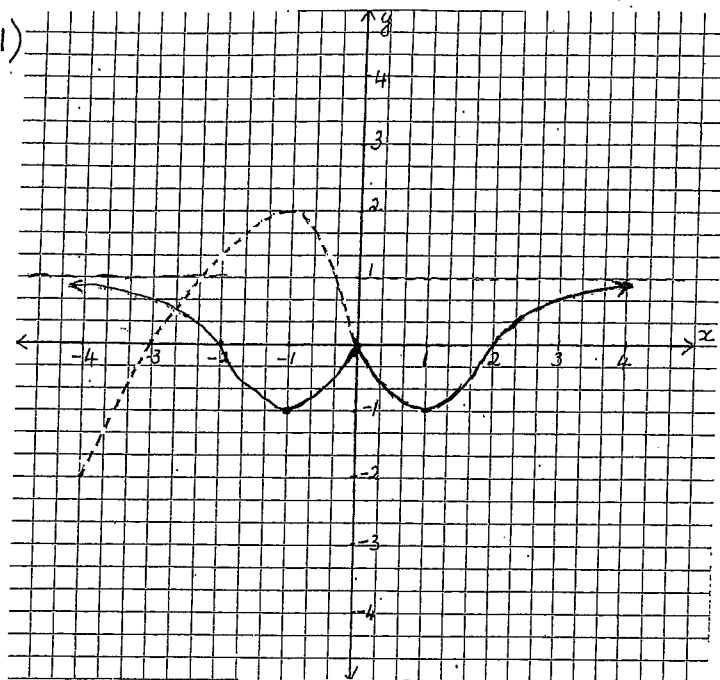
(ii) $y = [f(x)]^2$



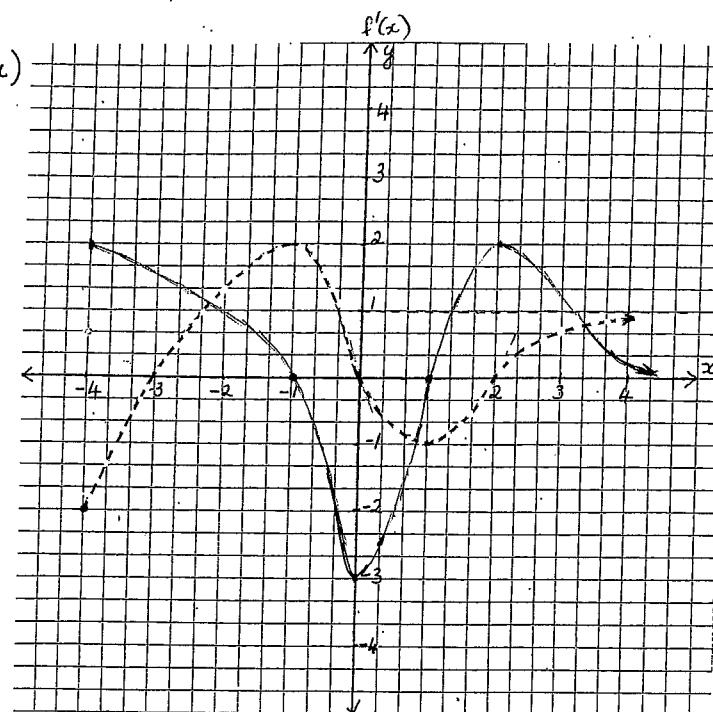
(iv) $y^2 = f(x)$



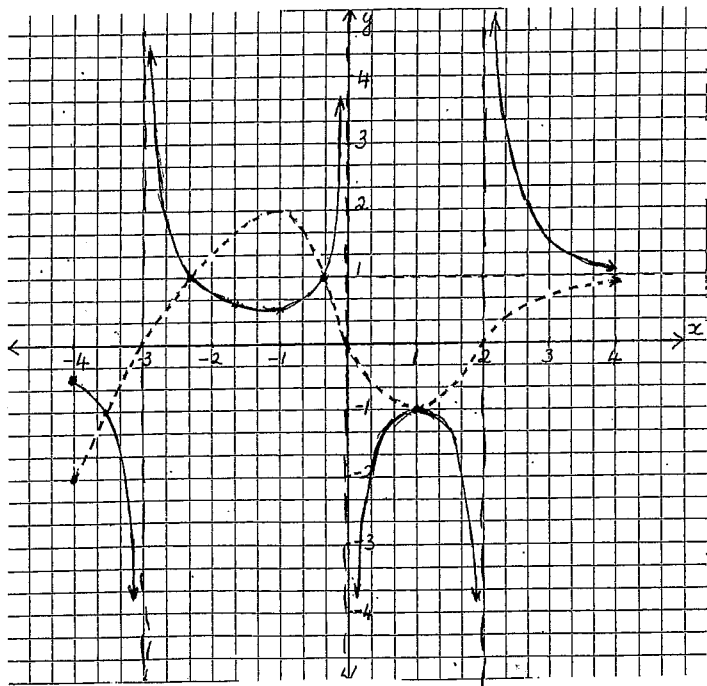
(vi) $y = f(|x|)$



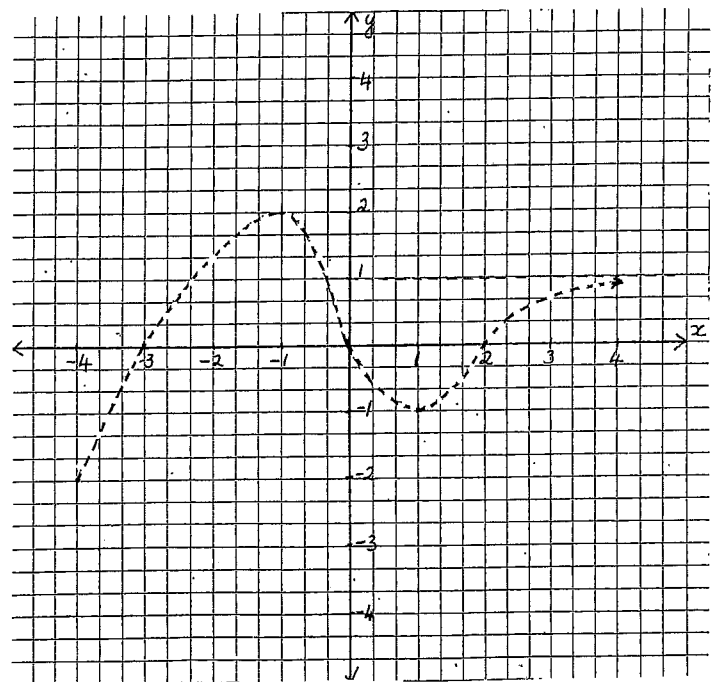
(vii) $y = f'(x)$



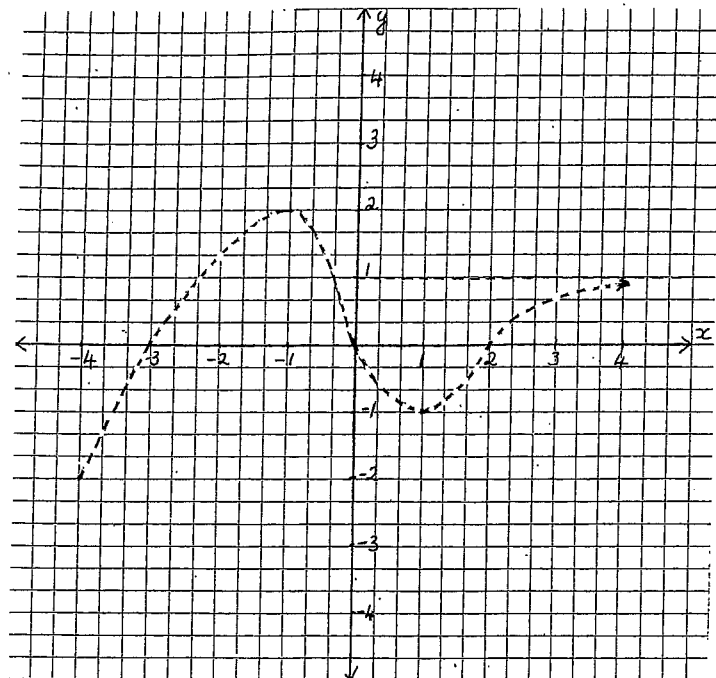
(i) $y = \frac{1}{f(x)}$



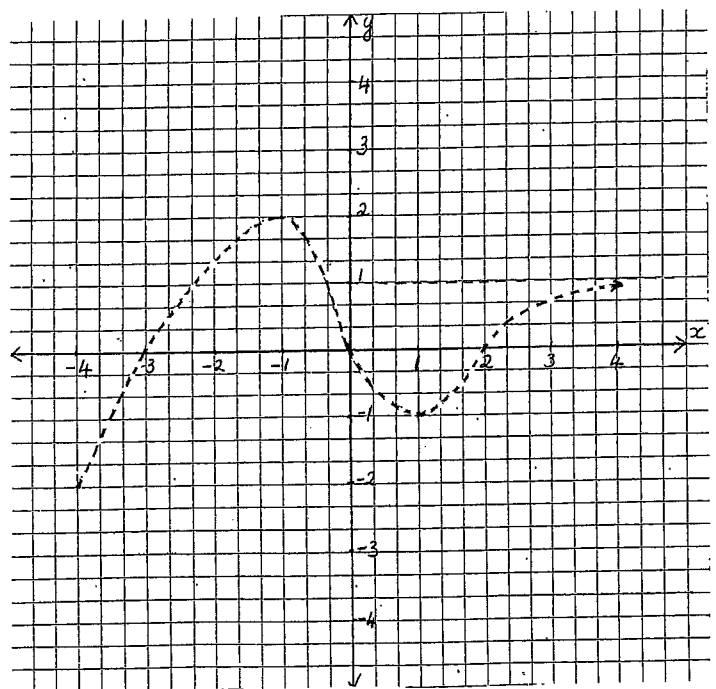
Spare
part:
equation:



part:
equation:



part:
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Question 2 (b)

$$\begin{aligned} \text{(i)} \quad & 2x^2 - xy + y^2 = 14 \\ & 4x - \left(y \cdot 1 + x \cdot \frac{dy}{dx}\right) + 2y \frac{dy}{dx} = 0 \\ & 4x - y + \frac{dy}{dx}(2y - x) = 0 \\ & \frac{dy}{dx} = \frac{y - 4x}{2y - x} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & \text{when } x = 1, y = -3 \quad \frac{dy}{dx} = \frac{-3 - 4 \times 1}{2 \times -3 - 1} \\ & = \frac{-3 - 4}{-6 - 1} \\ & = \frac{-7}{-7} \\ & = 1 \end{aligned}$$

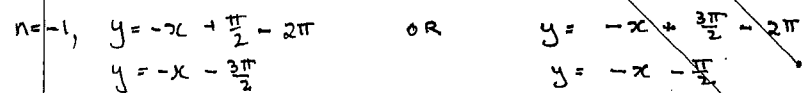
$$\begin{aligned} \text{(iii)} \quad & \text{Stationary Points occur when } \frac{dy}{dx} = 0 \\ & \text{ie } \frac{y - 4x}{2y - x} = 0, \quad 2y \neq x \end{aligned}$$

$$\begin{aligned} & \text{ie } y - 4x = 0 \\ & \text{ie all lie on } y = 4x \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad & \text{Stationary points must occur when } y = 4x \\ & \text{ie } 2x^2 - x(4x) + (4x)^2 = 14 \\ & 2x^2 - 4x^2 + 16x^2 = 14 \\ & 14x^2 = 14 \\ & x^2 = 1 \\ & x = \pm 1 \end{aligned}$$

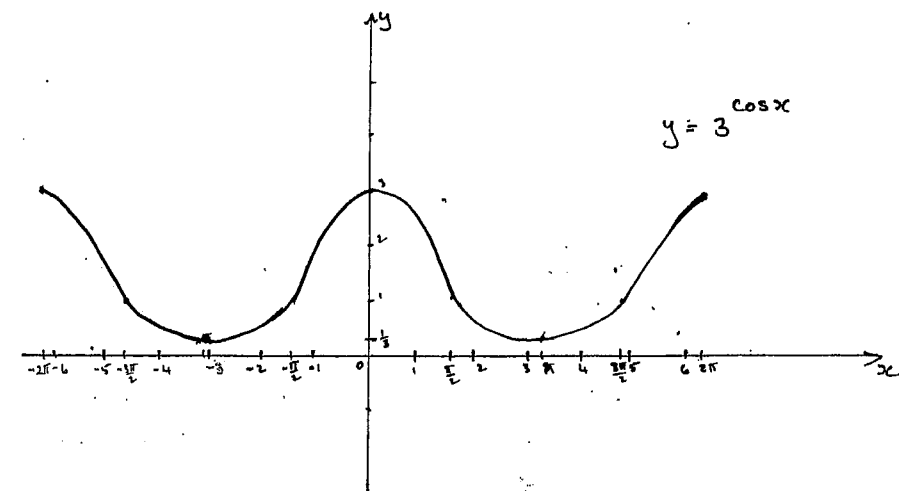
ie Stat Points are $(-1, -4)$ and $(1, 4)$

$$n=1, \quad y = -x + \frac{\pi}{2} + 2\pi \quad y = -x + \frac{3\pi}{2} + 2\pi$$



Graph continues as parallel lines separated by π units.

a) (i)



$$b) (i) \quad \frac{x^2}{9} - \frac{y^2}{7} = 1$$

Equation of tangent is

$$\frac{x_0 x}{9} - \frac{y_0 y}{7} = 1$$

Tangent crosses x -axis when $y=0$

$$\text{ie } \frac{x_0 x}{9} = 1 \Rightarrow x = \frac{9}{x_0}$$

$$\text{ie } T\left(\frac{9}{x_0}, 0\right)$$

(ii) From the definition of Hyperbola,

$$\frac{PS}{PM} = e \quad \text{and} \quad \frac{PS'}{PM'} = e$$

$$\therefore PS = e \cdot PM$$

$$PS' = e \cdot PM'$$

$$\therefore \frac{PS}{PS'} = \frac{e \cdot PM}{e \cdot PM'} = \frac{x_0 - \frac{3}{e}}{x_0 - -\frac{3}{e}} \quad \text{where } M \text{ is on } x = \frac{3}{e} \text{ and } M' \text{ is on } x = -\frac{3}{e}$$

$$= \frac{e x_0 - 3}{e x_0 + 3}$$

$$\text{Now also, } \frac{TS}{TS'} = \frac{3e - \frac{9}{x_0}}{\frac{9}{x_0} - -3e} \times x_0$$

$$= \frac{3e x_0 - 9}{9 + 3e x_0}$$

$$= \frac{3(e x_0 - 3)}{3(3 + e x_0)}$$

$$= \frac{e x_0 - 3}{3 + e x_0}$$

$$\text{Thus, } \frac{TS}{TS'} = \frac{PS}{PS'}$$

$$\begin{aligned} (iii) \quad |PS' - PS| &= |e \cdot PM' - e \cdot PM| \\ &= e |PM' - PM| \\ &= e \left| \left(x_0 - \frac{3}{e}\right) - \left(x_0 - -\frac{3}{e}\right) \right| \\ &= e \left| x_0 - \frac{3}{e} - x_0 + \frac{3}{e} \right| \\ &= e \left| -2 \times \frac{3}{e} \right| \\ &= |-6| \\ &= 6 \end{aligned}$$

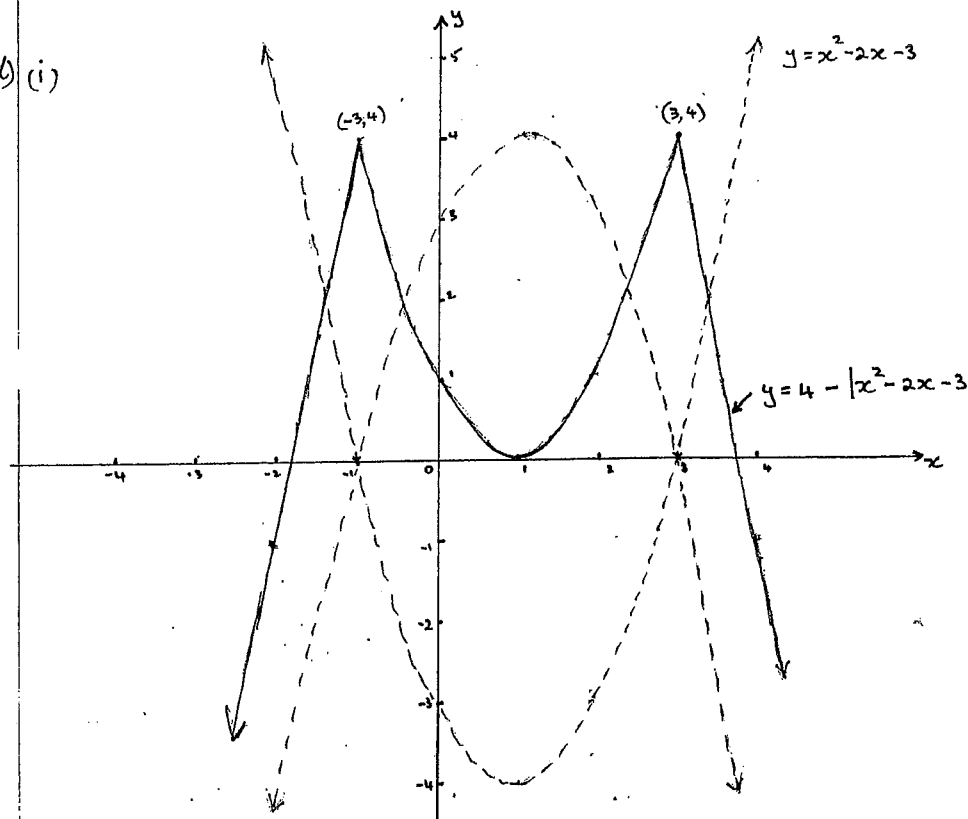
$$\text{ie } |PS' - PS| = 6$$

c) $y = mx + k$ is a tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ when Δ of quadratic formed on solving is ZERO.

$$\text{ie } \frac{x^2}{a^2} - \frac{(mx+k)^2}{b^2} = 1$$

$$\begin{aligned} x a^2 b^2: \quad & b^2 x^2 - a^2 (mx+k)^2 = a^2 b^2 \\ & b^2 x^2 - a^2 m^2 x^2 - 2a^2 m k x - a^2 k^2 = a^2 b^2 \\ & x^2 (b^2 - a^2 m^2) - 2a^2 m k x - a^2 (k^2 + b^2) = 0 \\ \therefore \Delta &= (-2a^2 m k)^2 - 4 \times (b^2 - a^2 m^2) \times -a^2 (k^2 + b^2) = 0 \\ &\text{ie } 4a^2 [a^2 m^2 k^2 + b^2 k^2 + b^4 - a^2 m^2 k^2 - a^2 b^2 m^2] = 0 \\ &\text{ie } b^2 (k^2 + b^2 - a^2 m^2) = 0 \\ &b^2 \neq 0, \quad m^2 a^2 - b^2 = k^2 \end{aligned}$$

d) (i)



$$y = (x-3)(x+1)$$

$$x=1, y = -2 \times 2 = -4$$

$$(1, -4) \text{ vertex}$$

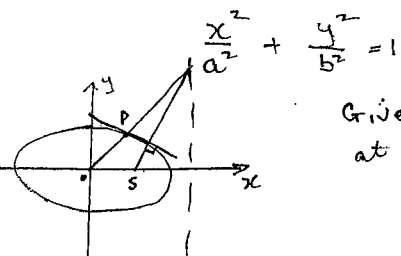
$$x=-2, y = -(x^2 - 2x - 3) = -(4 + 4 - 3) = -5$$

(ii)

$mx = 4 - |x^2 - 2x - 3|$
 Now, $y = mx$ passes through origin
 m is the gradient
 Thus for 4 distinct solutions

$$0 < m \leq \frac{4}{3}$$

e)



Given the equation of tangent at P is

$$y = \frac{-b \cos \theta}{a \sin \theta} \left(x - \frac{a}{\cos \theta} \right)$$

then, gradient of line perp. to tangent at P and through S is: $\frac{a \sin \theta}{b \cos \theta}$

$\therefore$ Equation of this line is:

$$y - 0 = \frac{a \sin \theta}{b \cos \theta} (x - ae)$$

$$y = \frac{a \sin \theta}{b \cos \theta} (x - ae) \quad \text{--- (1)}$$

Equation of line OP is $y = mx$

$$\text{i.e. } y = \frac{b \sin \theta - 0}{a \cos \theta - 0} x$$

$$y = \frac{b \sin \theta}{a \cos \theta} x \quad \text{--- (2)}$$

Point of intersection of lines (1) + (2) is:

$$\frac{a \sin \theta}{b \cos \theta} (x - ae) = \frac{b \sin \theta}{a \cos \theta} x$$

$$\frac{a}{b} (x - ae) = \frac{b}{a} x$$

$$a^2 (x - ae) = b^2 x$$

$$a^2 x - b^2 x = a^3 e$$

$$x = \frac{a^3 e}{a^2 - b^2}$$

$$\text{But } b^2 = a^2 (1 - e^2)$$

$$\therefore x = \frac{a^3 e}{a^2 - a^2 (1 - e^2)}$$

$$= \frac{ae}{1 - 1 + e^2}$$

$$x = \frac{a}{e}$$

i.e. lines meet on the directrix $x = \frac{a}{e}$